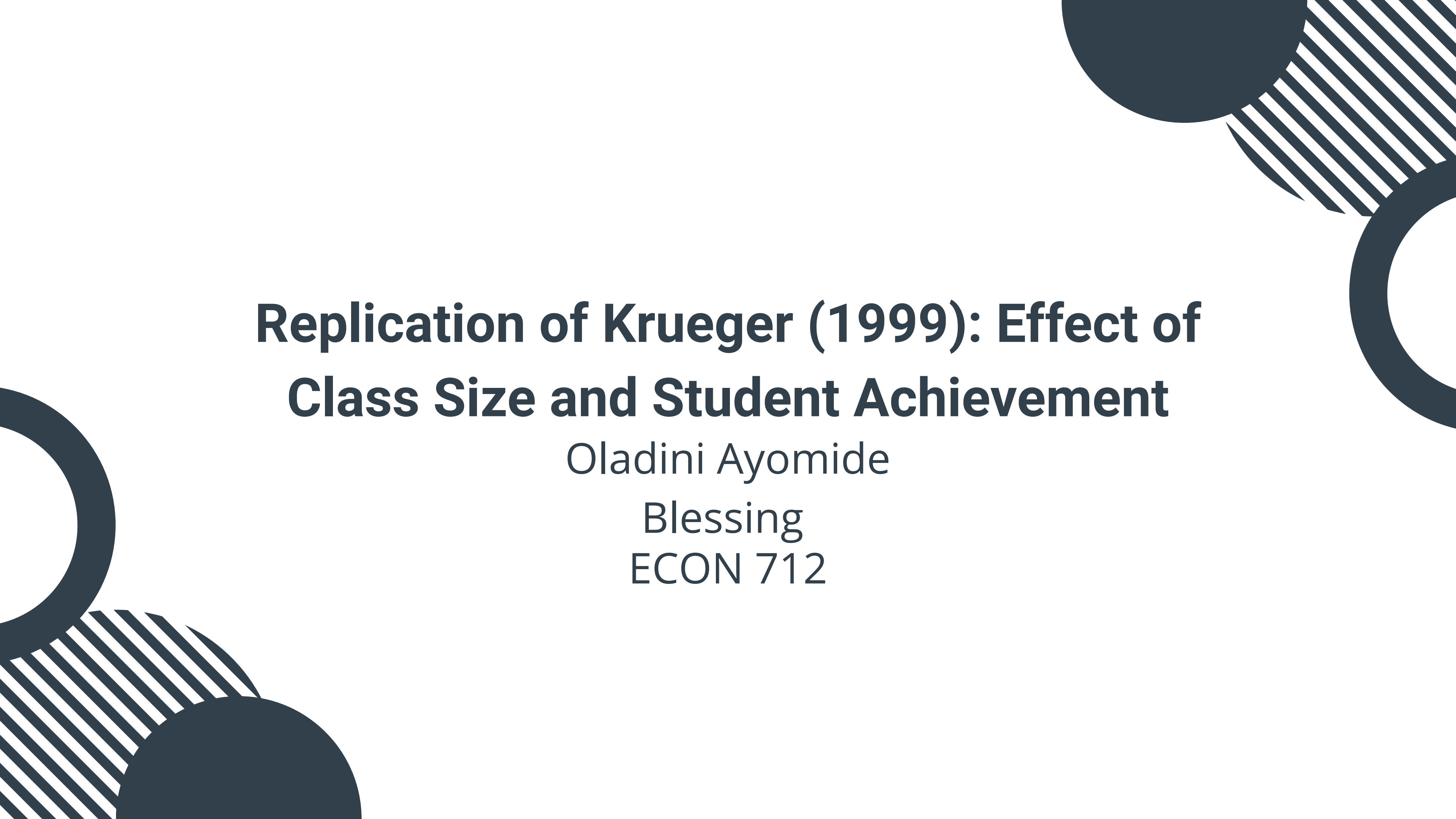


The slide features decorative elements in the corners: a solid dark blue circle in the top right, a circle with dark blue diagonal stripes in the top right, a solid dark blue circle in the bottom left, and a circle with dark blue diagonal stripes in the bottom left.

Replication of Krueger (1999): Effect of Class Size and Student Achievement

Oladini Ayomide

Blessing
ECON 712



Research Question

Does reducing class size improve student achievement? In other words, do smaller classes actually lead to better academic outcomes, especially in the early years of schooling?

This question matters because class size is one of the most debated education policies, and it is also very expensive to implement.

If smaller classes really improve student performance, then the cost might be justified.

Original Paper

This paper uses the Project STAR experiment (Tennessee), where students and teachers were randomly assigned to:

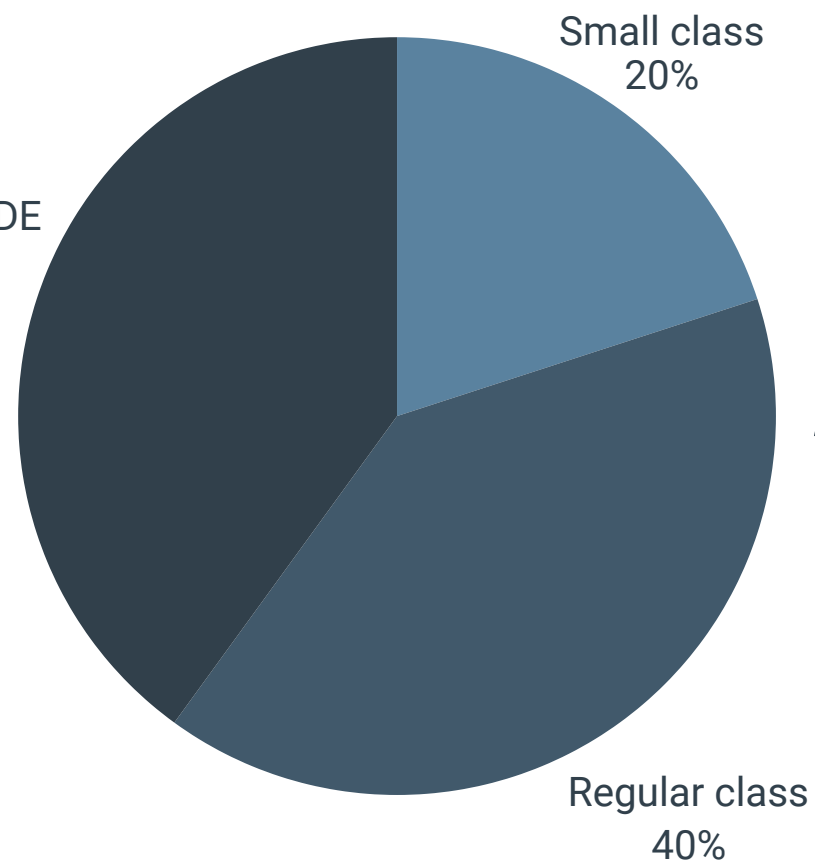
- *Small classes ($\approx 13\text{--}17$ students)*
- *Regular classes ($\approx 22\text{--}25$ students)*
- *Regular classes + Teacher AIDE ($\approx 22\text{--}25$) students*

The sample included 11,600 students from 80 schools, for a period of four years.

Findings:

Smaller classes do improve test scores, and Teacher aides have little to no effect compared to regular classes.

Class Size



Identification Strategy

Problem

Class size is usually endogenous. Selection bias, students not randomly assigned, causing omitted variable

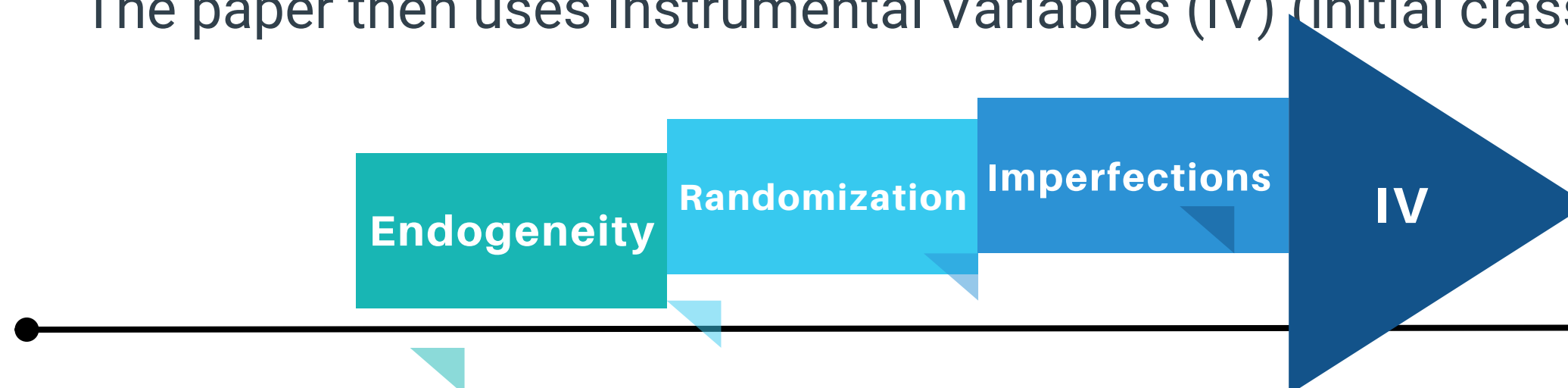
Strategy

Project STAR uses random assignment within schools to one of the three class type.

Complications

- Students switch classes
- Attrition over time

The paper then uses Instrumental Variables (IV) (initial class assignment)



DATA

Data source

Project STAR, the Tennessee Student/Teacher Achievement Ratio experiment.

Sample Structure

*11,600 students, 80 schools
Grades Kindergarten through third grade.
Random assignment within class type.*

Variables

*Test scores
(percentile-based),
using Math-SAT,
Reading-SAT, Word-SAT.
Treatment:
class type/class size
Controls:
age, gender, free
lunch status, teacher
race, teacher
gender, teacher
experience, and
teacher education.*

Challenges

- *Attrition across grades*
- *Class switching (treatment non-compliance)*
- *No baseline test scores*

TABLE 5
OLS AND REDUCED-FORM ESTIMATES OF EFFECT OF CLASS-SIZE ASSIGNMENT
ON
AVERAGE PERCENTILE OF STANFORD ACHIEVEMENT TEST

+

OLS: actual class size				Reduced form: initial class size				
A. Kindergarten								
Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small class	4.73	5.32	5.35	5.35	4.73	5.32	5.35	5.35
	(2.18)	(1.26)	(1.21)	(1.19)	(2.18)	(1.26)	(1.21)	(1.19)
Regular/aide class	-0.01	0.15	0.41	0.16	-0.01	0.15	0.41	0.16
	(2.21)	(1.13)	(1.09)	(1.07)	(2.21)	(1.13)	(1.09)	(1.07)
White/Asian	—	—	-1.17	-1.03	—	—	-1.17	-1.03
			(2.16)	(2.15)			(2.16)	(2.15)
Girl	—	—	4.46	4.38	—	—	4.46	4.38
			(0.63)	(0.63)			(0.63)	(0.63)
Free lunch	—	—	-14	-13.94	—	—	-14	-13.94
			(0.76)	(0.76)			(0.76)	(0.76)
White teacher	—	—	—	-0.59	—	—	—	-0.59
				(2.12)				(2.12)
Teacher experience	—	—	—	0.25	—	—	—	0.25
				(0.1)				(0.1)
Master's degree	—	—	—	-0.52	—	—	—	-0.52
				(1.06)				(1.06)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes

B. First grade

Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small class	8.6	8.31	7.19	6.73	7.46	6.95	5.76	5.49
	(1.9)	(1.15)	(1.19)	(1.22)	(1.7)	(1.09)	(1.13)	(1.14)
Regular/aide class	3.38	2.17	1.59	1.12	1.83	1.54	0.34	0.3
	(1.96)	(0.95)	(1.15)	(1.17)	(1.07)	(0.77)	(0.82)	(0.82)
White/Asian	—	—	-0.6	-0.18	—	—	-0.61	-0.17
			(1.67)	(1.64)			(1.71)	(1.66)
Girl	—	—	4.53	4.56	—	—	4.44	4.49
			(0.65)	(0.65)			(0.65)	(0.65)
Free lunch	—	—	-14.45	-14.57	—	—	-14.53	-14.66
			(1.04)	(1.04)			(1.05)	(1.05)
White teacher	—	—	—	-3.95	—	—	—	-4.08
				(2.14)				(2.16)
Male teacher	—	—	—	6.46	—	—	—	7.7
				(3.23)				(3.02)
Teacher experience	—	—	—	0.04	—	—	—	0.04
				(0.06)				(0.06)

Master's degree	—	—	—	2.68	—	—	—	3.45
				(1.56)				(1.4)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes

C. Second grade

Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small class	5.59	6.08	4.07	4.1	5.24	5.52	3.21	3.18
	(1.9)	(1.23)	(1.35)	(1.35)	(1.62)	(1.1)	(1.16)	(1.17)
Regular/aide class	1.08	1.38	1.04	0.99	-0.08	1.06	-0.12	-0.09
	(1.97)	(1.08)	(1.24)	(1.25)	(1.17)	(0.83)	(0.89)	(0.9)
White/Asian	—	—	-1.41	-0.78	—	—	-1.45	-0.83
			(1.73)	(1.72)			(1.74)	(1.75)
Girl	—	—	3.22	3.21	—	—	3.18	3.18
			(0.8)	(0.81)			(0.8)	(0.81)
Free lunch	—	—	-13.62	-13.66	—	—	-13.68	-13.71
			(0.98)	(0.98)			(1.01)	(1.01)
White teacher	—	—	—	-0.74	—	—	—	-0.64
				(1.87)				(1.89)
Male teacher	—	—	—	1.43	—	—	—	0.97
				(3.92)				(4.28)
Teacher experience	—	—	—	0.12	—	—	—	0.13
				(0.07)				(0.07)
Master's degree	—	—	—	-2.07	—	—	—	-2.38
				(2.07)				(2.05)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes

D. Third grade

Explanatory Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Small class	5.59	5.72	3.28	3.01	5.84	5.56	4.4	4.2
	(1.86)	(1.16)	(1.32)	(1.35)	(1.42)	(1.03)	(1.16)	(1.18)
Regular/aide class	-0.36	-0.1	-0.39	-0.55	-0.35	0.15	0.75	0.85
	(1.89)	(1.08)	(1.32)	(1.33)	(1.13)	(0.82)	(1.04)	(1.04)
White/Asian	—	—	-0.98	-0.55	—	—	-1.33	-0.94
			(1.86)	(1.83)			(1.9)	(1.86)
Girl	—	—	3.57	3.62	—	—	3.59	3.63
			(0.89)	(0.89)			(0.89)	(0.89)
Free lunch	—	—	-11.79	-11.42	—	—	-11.9	-11.52
			(1.12)	(1.13)			(1.13)	(1.14)

White teacher	—	—	—	0.63	—	—	—	0.12
				(2.08)				(2.08)
Male teacher	—	—	—	-7.11	—	—	—	-6.57
				(2.83)				(2.63)
Teacher experience	—	—	—	0.06	—	—	—	0.06
				(0.07)				(0.07)
School FE	No	Yes	Yes	Yes	No	Yes	Yes	Yes

TABLE 7

OLS AND 2SLS ESTIMATES OF EFFECT OF CLASS SIZE ON ACHIEVEMENT
DEPENDENT VARIABLE: AVERAGE PERCENTILE SCORE ON SAT

Grade	OLS	2SLS	Sample size
K	-0.63	-0.71	5,841
	(0.14)	(0.14)	
1	-0.82	-0.85	4,302
	(0.13)	(0.16)	
2	-0.42	-0.49	3,300
	(0.13)	(0.16)	
3	-0.36	-0.65	2,972
	(0.13)	(0.17)	

MAIN FINDINGS

- *Smaller classes improve student achievement*
- *Effects are strongest in early grades (Kindergarten and first grade)*
- *IV results confirm a negative effect of class size, suggesting a causal interpretation*
- *OLS understates the true causal effect of class size.*
- *Results are consistent across specifications*
- *Class size reduction has a meaningful educational impact*
- *Effects are economically meaningful, not just statistically significant*

Limitations and Proposed Improvements

Limitations

- *Attrition across grades*
 - *Class switching (non-compliance)*
 - *No baseline test scores*
-

Proposed Improvements

- *Include baseline achievement measures*
- *Track the long-run outcomes (education, labour market)*
- *Improve handling of attrition*



Conclusion

In conclusion, the replication confirms that smaller classes improve student achievement, especially in the early grades.

Using the experimental design and instrumental variables approach, the results support a causal interpretation.

The findings are robust across different specifications

From a policy perspective, this suggests that reducing class size can be an effective way to improve educational outcomes, particularly in early education.

Overall, the paper highlights the importance of early educational interventions.





THANK YOU